Seyed Alireza Mousavian Anaraki

PhD in Data Science, University of Rome Tor Vergata | Visiting Researcher, University of Sheffield
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Google Scholar | ResearchGate | LinkedIn

EDUCATION

PhD Student in Data Science — Nov. 2023 – Oct. 2026 (Expected*)
University of Rome Tor Vergata, Department of Enterprise Engineering — Rome, Italy

- **Research Focus**: **NLP for sustainability (ESG) analysis**, large-scale processing of sustainability reports, semantic alignment and weak-supervision techniques, and LLMs for structure, consistency, and credibility assessment
- *Thesis defense scheduled for April–May 2027
- **Supervisors**: Profs. Danilo Croce (croce@dii.uniroma2.it) and Roberto Basili (basili@dii.uniroma2.it)

Master's Degree in Engineering Management — Sept. 2019 – Nov. 2021
Iran University of Science and Technology, Department of Industrial Engineering — Tehran, Iran

- **GPA: 19.33/20 — Ranked 1st** student in Master's program
- **Thesis**: Development and implementation of data mining methods by creating strong, balanced, and interpretable clusters to adopt appropriate policies
- **Supervisor**: Dr. Abdorrahman Haeri (ahaeri@iust.ac.ir)

Bachelor's Degree in Industrial Engineering — Sept. 2015 – Aug. 2019
Kashan University, Department of Industrial Engineering — Kashan, Iran

- **GPA: 18.62/20 — Ranked 1st** student in Bachelor's program

RESEARCH EXPERIENCE

Visiting Researcher — Apr. 2026 – Aug. 2026
University of Sheffield, School of Computer Science — Sheffield, UK

- Investigated **credibility bias in ESG language models**, focusing on how geographic, institutional, and linguistic factors influence automated sustainability assessment
- **Supervisor**: Dr. Nafise Sadat Moosavi (n.s.moosavi@sheffield.ac.uk)

Researcher and Data Analyst — Nov. 2020 – Aug. 2021
Iran's National Elites Foundation (INEF) — Tehran, Iran

- Served as **data-driven reliability-centered maintenance researcher** within the data analytics research group, improving maintenance of critical energy infrastructure
- **Supervisor**: Dr. Alireza Fereidunian (fereidunian@kntu.ac.ir)

Research Assistant — Sept. 2019 – Nov. 2021
Iran University of Science and Technology — Tehran, Iran

- Developed and implemented **data mining methods** across multiple domains with an emphasis on data-driven decision-making
- **Supervisor**: Dr. Abdorrahman Haeri (ahaeri@iust.ac.ir)

TEACHING EXPERIENCE

Teaching Assistant — Linear Algebra — Sept. 2017 – Feb. 2018
Kashan University — Kashan, Iran

- **Supervisor**: Dr. Mohammad Taghi Rezvan (rezvan@kashanu.ac.ir)

PUBLICATIONS

Journal Papers

- Anaraki, S. A. M., Croce, D., & Basili, R. (2025). Large language models for sustainability reporting: A systematic review and research agenda. *Sustainable Futures*, 10, 101494. doi.org/10.1016/j.sftr.2025.101494
- Fallahi, A., Mousavian Anaraki, S. A., Mokhtari, H., & Niaki, S. T. A. (2022). Blood plasma supply chain planning to respond COVID-19 pandemic: a case study. *Environment, Development and Sustainability*, 1-52. doi.org/10.1007/s10668-022-02793-7
- Anaraki, S. A. M., & Haeri, A. (2022). Soft and hard hybrid balanced clustering with innovative qualitative balancing approach. *Information Sciences*. doi.org/10.1016/j.ins.2022.09.044
- Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2022). Generating balanced and strong clusters based on balance-constrained clustering approach (strong balance-constrained clustering) for improving ensemble classifier performance. *Neural Computing and Applications*, 1-17. doi.org/10.1007/s00521-022-07595-6
- Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2021). A hybrid reciprocal model of PCA and K-means with an innovative approach of considering sub-datasets for the improvement of K-means initialization and step-by-step labeling to create clusters with high interpretability. *Pattern Analysis and Applications*, 24(3), 1387–1402. doi.org/10.1007/s10044-021-00977-x
- Mousavian, S. A., Haeri, A., & Moslehi, F. (2021). Providing a Hybrid Clustering Method as an Auxiliary System in Automatic Labeling to Divide Employee into Different Levels of Productivity and their Retention. *Iranian Journal of Management Studies*. doi:10.22059/ijms.2021.299705.674004

Conference Papers

- Mousavian Anaraki, S. A., Croce, D., Costa, R., Tiburzi, L., Calabrese, A., & Basili, R. (2026). Unsupervised GRI-TCFD Alignment with LLM-Assisted Validation for Climate Disclosure and Greenwashing Risk Analysis. In *Proceedings of the 2nd Workshop on Ecology, Environment, and Natural Language Processing* (pp. 15–25). ELRA. doi.org/10.63317/4pebkuscwua5
- Borazio, F., Anaraki, S. A. M., Rai, S. I., Croce, D., & Basili, R. (2026). UniTor at PFB: Constrained Agentic Prompting and Self-Consistency for Financial Multi-Choice QA. In *EVALITA*.
- Anaraki, S. A. M., Croce, D., & Basili, R. (2025, September). Automatic GRI-SDG Annotation and LLM-Based Filtering for Sustainability Reports. In *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)* (pp. 775-784). https://aclanthology.org/2025.clicit-1.73/
- Anaraki, S. A. M., Croce, D., & Basili, R. (2025, July). Unsupervised Sustainability Report Labeling based on the integration of the GRI and SDG standards. In *Proceedings of the Fourth Workshop on NLP for Positive Impact (NLP4PI)* (pp. 151-162). aclanthology.org/2025.nlp4pi-1.13

HONORS AND AWARDS

- **Outstanding Paper Award**, Workshop on NLP for Positive Impact (NLP4PI @ ACL 2025) — July 2025
- **Ranked 1st** student in Master's program — July 2020
- Eligible to continue **directly to the Master's Program** without the National University Entrance Exam – Sept. 2019
- **Ranked 1st** student in Bachelor's program — July 2018

RESEARCH INTERESTS

- Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL)
- Information Extraction and Natural Language Processing (NLP) Applications
- Interpretability and Analysis of NLP Models
- NLP and LLMs for Environmental, Social, and Governance (ESG) Sustainability Reporting
- AI-Driven Decision-Making in Management Applications
- Operations Research Applications

SKILLS

Programming : Python, R, SQL Server
Data Visualization / BI : Tableau, Power BI
Engineering Software : GAMS, COMFAR, Expert Choice, ALDEP, VOSviewer
Languages : English (Upper Intermediate), Persian (Native)